1 Problem Description

Given a 45x45 square, how many landscape rectangles are there?

2 Problem Solution

With the 4x4 there are 35. This is because there are

$$\begin{aligned} 12 &: 2 \times 1, \\ 8 &: 3 \times 1, \\ 4 &: 4 \times 1, \\ 6 &: 3 \times 2, \\ 3 &: 4 \times 2, \\ 2 &: 4 \times 3 \\ &=35 \end{aligned}$$

Or formatted differently (think about the overhang after shifting a rectangle each time, it will be the number of rectangle columns - 1 and similarly for rows so if you subtract that from the total unit shifts you have the number of combos)

$$\begin{aligned} (4-1) \times (4-0) &: 2 \times 1, \\ (4-2) \times (4-0) &: 3 \times 1, \\ (4-3) \times (4-0) &: 4 \times 1, \\ (4-2) \times (4-1) &: 3 \times 2, \\ (4-3) \times (4-1) &: 4 \times 2, \\ (4-3) \times (4-2) &: 4 \times 3, \\ (4-(col-1)) \times (4-(row-1)) &: c \times r \end{aligned}$$

So for 45 it would be

$$\sum_{row=1}^{45} \sum_{col=row+1}^{45} (45 - (col - 1)) \times (45 - (row - 1))$$

Sub into desmos and get: 519915